



Syllabus

Software Engineering Department

Course Name	Artificial Intelligence	
Course No.	3502193	
Lecturers' Name	Dr. Braha Riva Shalom	riva.shalom@gmail.com
Teaching Assistant		
Year	2025-2026	תשפ"ו
Weekly Hours	3	
Credits	3	
Prerequisite	Data Structures 3502810, Mathenatic Logic 3500851	

Course Summary
The course will introduce students with the main research fields of AI. The course aims at providing students with practical knowledge of transferring real life problems into a formal description, understood by the computer. The course aims at providing students with the knowledge of the basic algorithms in different disciplines of AI.

LEARNING METHODS	The course is a blended course, the learning process includes frontal lectures using presentations, short movies, class technical exercises, class discussions and self-study guided by pre-class assygnments.
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Learning Outcomes
1. Students will understand the contribution of AI today.
2. Students will know the main disciplines in AI.
3. Students will understand how to use a variety of AI algorithms from different disciplines.
4. Students will be able to convert a real life problem to a formalized problem.
5. Students will be able to identify and implement the appropriate AI algorithm required in their final project.

Course Requirements and Grading			
Course Requirements	At least 60 in the final exam		
Grading	Pre-Class Assignment	10%	10 out of 12 are required
	After-Class Assignments	10%	10 out of 12 ex. are required
	Test	80%	

Class Attendance	Mandatory. According to Shenkar's regulations
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Course Plan- List of Topics		
	Topic	Reading/Assignment
1	Introduction to Artificial Intelligence. Introduction to Agents.	Chapter 1,2
2	Defining formal problems. Solving Problems by Searching. Basic Search algorithms: BFS, DFS.	Chapter 3/ Ex. 1
3	Advanced Search Algorithms. Iterative Deepening, Bi directional Search, Uniform Cost Search	Chapter3/ Ex. 2
4	Informed Search: Heuristic Functions, Greedy Search	Chapter 4/ Ex. 3
5	More informed search : A* algorithm, Local Search, Genetic Algorithms.	Chapter 4/ Ex. 4
6	Constraint Satisfaction Problems.	Chapter 5/ Ex. 5
7	AI in Games. Minmax Algorithm, Alpha Beta Pruning	Chapter 6/ Ex. 6
8	Logical Agents: Knowledge representation. Propositional Logic. DPLL.	Chapter 7/ Ex. 7
9	Propositional Logic Resolution, Forward/Backward Chaining	Chapter 7/ Ex. 8
10	First Order Logic, necessity and rules. First Order Resolution.	Chapter 8/ Ex. 9
11	First Order Logic Forward/Backward Chaining.	Chapter 9/ Ex. 10
12	Learning From Observations: Decision Trees.	Chapter 18/ Ex. 11

13	Planning	Chapter 11/ Ex.12 Chapter 10
14	Course Wrap-up	

Bibliography	
Artificial Intelligence: A modern Approach (Fourth Edition) by Stuart Russell and Peter Norvig , 2020, Pearson.	